

Solving Fermat's Last Theorem: Andrew Wiles' Breakthrough After Over 350 Years

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1 Pierre de Fermat

Pierre de Fermat was a famous 17th-century French mathematician whose nickname is “the founder of the modern theory of numbers.” He is often known for his early developments that led to modern calculus and his contribution to the early progress of probability theory. Being inspired by the “Arithmetica” of the mathematician Diophantus, he continued to discover various new patterns in numbers which led to the creation of many new conjectures and theorems, but the most unique one was his last theorem.

2 Fermat's Last Theorem

Fermat's Last Theorem was called his *pièce de résistance*, a conjecture he was working on but left unproven until the day of his death. Fermat's mathematical work was usually communicated through letters to his colleagues or friends but with little or no proof. Even though he claimed to have proved all the arithmetic theorems, there are only a few records left to actually show it. His unfinished last theorem puzzled mathematicians for over 350 years. The theorem was originally described in a scribbled note in the margin of his copy of Diophantus' "Arithmetica." Because there was no proof for his theorems, many mathematicians began to doubt his claims, especially since the conjecture has been found to be one of the hardest mathematical problems in the world to prove.

Fermat's Last Theorem makes the statement that there are no natural numbers $(1, 2, 3, \dots)$ a, b , and c such that $a^n + b^n = c^n$, in which n is a natural number greater than 2. For example, if $n = 3$, Fermat's Last Theorem states that no natural numbers x, y , and z exist such that $x^3 + y^3 = z^3$.

3 Proof for Fermat's Last Theorem

Countless mathematicians were confused about how to finish his proof. There were proofs for many different parts of the theorem; many for specific values of n were devised. For example, Fermat himself did a proof of another theorem that effectively solved the case for $n = 4$, and by 1993, with the help of computers, it was confirmed for all prime numbers $n < 4,000,000$. By that time, mathematicians had discovered that proving a special case of a result from algebraic geometry and number theory known as the Shimura-Taniyama-Weil conjecture would be equivalent to proving Fermat's Last Theorem.

However, one mathematician in particular named Andrew Wiles broke new ground. He was an English mathematician who had been interested in the theorem since he was 10 years old and finally presented the proof of the Shimura-Taniyama-Weil conjecture in 1993. When revised, there was an error that was found in this proof, but with help from his former student Richard Taylor, Wiles finally devised a proof of Fermat's Last Theorem. His work of the proof was published in 1995 in the journal *Annals of Mathematics*, and the Norwegian Academy of Science and Letters awarded him with the 2016 Abel Prize "for his stunning proof of Fermat's Last Theorem by way of the modularity conjecture for semistable elliptic curves, opening a new era in number theory."

4 Wiles' Proof

Wiles' proof has three main parts to it. The first part is setting up for the proof. We start by assuming (for the sake of contradiction) that Fermat's Last Theorem is incorrect. That would mean there is at least one non-zero solution (a, b, c, n) (with all numbers rational and $n > 2$ and prime) to $a^n + b^n = c^n$. Then we use Ribet's theorem (using Frey and Serre's work) to show that we

can create a semi-stable elliptic curve E using the numbers $(a, b, c,$ and $n)$, which is never modular. If we can prove that all such elliptic curves will be modular (meaning that they match a modular form), then we have our contradiction and have proved our assumption (that such a set of numbers exists) was wrong. If the assumption is wrong, that means no such numbers exist, which proves Fermat's Last Theorem is correct.

The second part is the Modularity Lifting theorem where Wiles aims to first prove a result about these Galois representations of elliptic curves $\rho(E, p)$ for any prime $p > 3$, that he will use later: that if a semi-stable elliptic curve E has a Galois representation $\rho(E, p)$ that is modular, the elliptic curve itself must be modular. By proving this, it makes counting and matching easier and proves the representation is modular. Then Wiles used proof by induction and a class number formula ("CNF") to show that once the hypothesis for one elliptic curve is proved, it can automatically be extended to be proven for all subsequent elliptic curves. This shows a key point about Galois representations which is that if the geometric Galois representation $\rho(E, p)$ of a semi-stable elliptic curve E is irreducible and modular (for some prime number $p > 2$), then subject to some technical conditions, E is modular. This is Wiles' lifting theorem which he continues to part 3 where he finally proves that all semistable elliptic curves are modular using the modularity lifting theorem. The final result proved that whether or not $\rho(E, 3)$ is irreducible, E (which could be any semi-stable elliptic curve) will always be modular. This means that all semistable elliptic curves must be modular.

June 23 marks the date where Andrew Wiles proved Fermat's Last Theorem, the most famous math problem at the time. After Wiles had completed this great achievement, there were more ideas that talked about a new golden age of mathematics in number theory, which the Last Theorem was known for. While his proof was published, there were still many who questioned him and his answers. One such misunderstandings was in Fermat's Last Theorem which said, "If n is any positive integer greater than 2, then it is impossible to find three positive numbers a, b and c such that $a^n + b^n = c^n$ ". This idea contrasted with what happens to the Pythagorean triples when $n = 2$ like $3^2 + 4^2 = 5^2$, where the rule does work and isn't impossible. Mathematicians tried to explain this contrast to the best of their ability, but it just led to more questions. Eventually, a computer scientist provided a logical analysis of Wiles' proof, and this automated proof of verification helped give the mathematicians some solutions to their questions. The verifications involve reformulating the proof as a series of statements, each written in a language understood by the computer, which then confirms every step and stamps the constitutional certification. This method is usually applied to long and extremely difficult proofs.

This method helped them figure out that if there is a contrary to Fermat's Last Theorem where there is a triple of positive integers $a, b,$ and c that $a^p + b^p = c^p$, for an odd number p where this always stood true. Otherwise, $a, b,$ and c could be rearranged into a new equation called an elliptic curve. The elliptic curve is shown into a Galois representation which consists of an infinite collection of equations that list the properties that were expected to be impossible. Every Galois representation should be assigned by a modular form. Yutaka Taniyama and Goro Shimura are

credited with the proof for this solution. Wiles had only proved a small part of this modularity conjecture, and it was finally established by other mathematicians like Christophe Breuil, Brian Conrad, etc.

5 Conclusion

Fermat's Last Theorem is definitely one of the most challenging math problems in history and was only solved because of the hard work of countless mathematicians starting with Fermat, then Andrew Wiles, and leading to mathematicians who try to understand it even today. One application of Fermat's Last Theorem is found in this question: How many ordered triplets exist when you define a primitive inverse-cubed triplet (x, y, z) where the $\gcd(x, y, z) = 1$ and $\frac{1}{x^3} + \frac{1}{y^3} = \frac{1}{z^3}$? We know the answer is that no such (x, y, z) exists because by Fermat's Last Theorem, the only solution occurs when $x^2y^2z^2 = 0$. More applications of this important theorem will no doubt be discovered for years to come.

6 Problems for Further Exploration

Problem 1: What is the number of distinct positive integers n such that $n + 3^2$ and $n^2 + 3^3$ are both perfect cubes?

Problem 2: Let n be a positive integer which makes $(4n^2 + 6n + 3)\frac{n}{3}$ a triangular number. How many solutions are there for n ?

Problem 3: Find the number of positive integers n such that there exists an integer $m \neq \frac{n}{2}$ such that $\frac{n^3 - 2m^3}{6m}$ is a perfect square.

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